

SANSKRITI BINANI

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Electrical engineer specializing in **end-to-end PCB design** (Altium Designer, KiCad) from **schematic capture** and **multi-layer / rigid-flex layout** through **bring-up, validation**, and manufacturing hand-off. Hands-on with **DC-DC power conversion** (buck, boost, LDO), power-rail and **RF layout, stackup and constraint management**, and circuit simulation (PSPice, TI WEBENCH). Delivered functional, test-validated prototypes across startup and research environments.

EDUCATION

University of Pennsylvania

Master of Science in Electrical Engineering

May 2026

Philadelphia, PA

- **Coursework:** Digital Integrated Circuits, Embedded Systems, Networks & Protocols, IoT Edge Computing, Signal Processing

SKILLS

EDA & Simulation : Altium Designer, KiCad, Solidworks, TI WEBENCH, MATLAB, NI LabVIEW

MCUs & Hardware: ESP32, nRF54L15, SiWx917Y, ATmega328PB, SAMW25, Raspberry Pi

Lab & Test : Oscilloscope, Logic Analyzer, DMM, JTAG/SWD, SMD Soldering, Reflow Oven

Protocols : SPI, I2C, UART, CAN, BLE, MQTT, LoRa (915 MHz), Wi-Fi (2.4 GHz)

Programming : C/C++, Python, RTOS (FreeRTOS), Git, Linux

EXPERIENCE

Rainmaker Technologies

Jun 2026 – Present

Electrical Engineer Intern

Los Angeles, CA

- **Redesigned** the next-generation airborne PTH sensor board as a **4 layered rigid-flex PCB** in **Altium Designer**, re-architecting the front end to reduce board thermal mass and cut τ_{63} response time from 2-4 s to under 1 s reaching parity with the iMet-4 radiosonde reference standard. Validated on a **first-pass build with no respin**.
- **Calibrated and characterized** each board's capacitive RH sensor individually across temperature and humidity setpoints in environmental chambers, deriving per-unit calibration coefficients and qualifying all 5 units for production release.
- **Owned** design and build of the team's first **motor test bench** using custom **relay-matrix PCB, 4-wire Kelvin** routing, measuring phase resistance to **1 m Ω accuracy** and recovering **~6 motor sets** from scrap via a **15s** automated screen before production.

University of Pennsylvania

Jan 2026 – May 2026

Teaching Assistant – IoT Edge Computing

Philadelphia, PA

- **Mentored** 10+ project teams, diagnosing wireless connectivity, firmware, and PCB-level failures using oscilloscopes and logic analyzers; authored validation workflows and troubleshooting documentation, improving project success rates by 40%.
- **Led** weekly hardware bring-up sessions covering PCB layout practices (impedance control, SMD soldering), signal integrity validation, and SiWx917Y Wi-Fi module configuration, reducing debug cycles and improving first-pass hardware success.

FilterFox Inc.

May 2025 – Dec 2025

Hardware Engineering Intern

Philadelphia, PA

- **Led** electrical integration and system bring-up of an ESP32-based IoT product, owning **schematic capture, PCB layout (Altium Designer), RF communication, power, and cloud connectivity** end-to-end.
- **Designed and routed RF traces** for dual wireless communication (**LoRa at 915 MHz, Wi-Fi at 2.4 GHz**), optimizing antenna matching and layout to ensure reliable signal integrity and range.
- **Architected** an **event-driven power management** scheme to trigger wake events, extending battery life by **65–70%**.
- **Coordinated** with PCB vendors on design reviews and manufacturing constraints and managed component inventory to mitigate supply-chain delays, keeping hardware prototyping on schedule.

GRASP Robotics Lab – UPenn

Jan 2025 – Aug 2025

Researcher

Philadelphia, PA

- **Designed** a **4-layer PCB** (EPC901 image-sensor breakout board) and **developed firmware** on **nRF54L15** using timer-triggered **SAADC, DPPI-based hardware triggering**, and double-buffered sampling for continuous **high-speed data capture**.

PROJECTS

FoldEasy | Automated Clothes-Folding Machine

- **Architected** an automated clothes-folding system on a **4-layer PCB** using an Atmel SAMW25 MCU, integrating **USB-C input, Li-ion BMS, DC-DC power regulation**, and **TI WEBENCH** for component selection to ensure reliable, safe operation.
- **Engineered** FreeRTOS-based firmware with semaphore-protected resource management, an interactive CLI, and a secure bootloader enabling robust over-the-air firmware updates (OTAUFU).
- **Built** an Azure IoT-integrated MQTT pipeline publishing system health and processing metrics at 30 s intervals, with real-time visualization on a Node-RED dashboard via WINC1500 Wi-Fi.

R.E.F.R.E.S.H. | Wearable Cooling Glove for F1 Drivers

- **Developed** a wearable cooling glove integrating GSR and MAXREFDES117 SpO2 sensors over **ADC and I2C** for real-time physiological monitoring, with a **PWM-controlled** cooling loop driven by ADC-mapped sweat levels.
- **Streamed** real-time sensor telemetry over an ESP32 to the Blynk IoT platform and trained a **Decision Tree** model on GSR data, achieving **98%** accuracy in hydration-level prediction.